

Predicting the Grade of Clear Cell Renal Cell Carcinoma from CT Images Using Random Subspace-KNN and Random Forest Classifiers

Ahmet Karagoz

Biomedical Engineering Institute
Bogazici University
Istanbul, Turkey
ahmet.karagoz@boun.edu.tr

Albert Guvenis

Biomedical Engineering Institute
Bogazici University
Istanbul, Turkey
guvenis@boun.edu.tr

Abstract—Accurate and non-invasive determination of the International Society of Urological Pathology (ISUP) based tumor grade is important for the effective management of patients with clear cell renal cell carcinoma (cc-RCC). In this study, the radiomic analysis of 3D computed tomography (CT) images are used to determine ISUP grades of cc-RCC patients by exploring machine learning (ML) methods that can address small ISUP grade image datasets. 143 cc-RCC patient studies from The Cancer Imaging Archive (TCIA) USA were used in the study. 1133 radiomic features were extracted from the normalized 3D segmented CT images. Correlation coefficient analysis, Random Forest feature importance analysis and backward elimination methods were used consecutively to reduce the number of features. 15 out of 1133 features were selected. A k -nearest neighbors (KNN) classifier with random subspaces and a Random Forest classifier were implemented. Model performances were evaluated independently on the unused 20% of the original imbalanced data. ISUP grades were predicted by a KNN classifier under random subspaces with an accuracy of 90% and area under the curve (AUC) of 0.88 using the test data. Grades were predicted by a Random Forest classifier with an accuracy of 83% and AUC of 0.80 using the test data. In conclusion, ensemble classifiers can be used to predict the ISUP grade of cc-RCC tumors from CT images with sufficient reliability. Larger datasets and new types of features are currently being investigated.

Keywords—cc-RCC, computed tomography, machine learning, radiomics, WHO/ISUP grade

I. INTRODUCTION

Renal cell carcinoma (RCC) constitutes 85% to 90% of all the kidney malignancies. It has an estimated mortality rate of 140,000 per year. About 3.7% of patients, who are newly diagnosed as cancer, are RCC [1].

There are several subtypes of RCC. More than nine in ten of the RCCs are clear cell renal cell carcinomas (cc-RCC), papillary renal cell carcinomas (pRCC), and chromophobe renal cell carcinomas (crRCC). The most common RCC subtype is cc-RCC which represents 75% of the all cases [2].

Grading and staging are important steps for cancer diagnosis, treatment, and prognosis [3]. A tumor grade indicates aggressiveness that is, how quickly it is expected to spread and conventionally is determined by a microscope examination of a biopsy sample. Several tumor grading methods have been recommended for RCC tumors [4]. One of the most common systems was the Fuhrman grading system that has been widely used at an international level.

Because of intra-observer and inter-observer reproducibility issues in the 4-tiered Fuhrman grading system, a novel World Health Organization (WHO)/International Society of Urological Pathology (ISUP) grading system was developed and presented during the 2012 ISUP Vancouver conference. This new grading system was recommended by WHO in 2016. This system is also composed of four grades. The highest-grade cell is used to assign a tumor grade [5].

The gold standard for preoperative tumor grading involves the use of invasive techniques such as fine-needle aspiration biopsy (FNAB), percutaneous kidney biopsy (PKB) or surgical excision. However, invasive methods are known to have serious drawbacks such as hemorrhages, infection, tumor spreading, contamination and a lack of information on the whole tumor and metastatic tumors [6,7]. Therefore, it is important to find less invasive methodologies to predict tumor grades accurately.

Recently, radiomic features obtained from medical imaging systems such as computed tomography (CT), magnetic resonance imaging (MRI) and positron emission tomography (PET) have been used to produce precise and robust clinical decision support systems (CDSS) for cancer detection, diagnosis, or prognosis [8]-[10]. Several studies have focused on predicting prognosis or some tumor characteristics, but few studies exist that aim to predict ISUP tumor grade from CT images. In studies [11]-[14], radiomics were utilized for differentiating 2-tiered Fuhrman or WHO/ISUP grades that were low and high (low stands for grades 1 and 2, and high stands for grades 3 and 4). These studies used mostly relatively larger datasets to build machine learning based prediction models.

ISUP grade datasets obtained from imaging equipment can be small. In the present study, we aimed to tackle this challenge by creating and testing two ensemble learning algorithms using radiomic features obtained from a small dataset of three dimensional (3D) CT images. The goal was to predict the correct WHO/ISUP grade of cc-RCC patients preoperatively and noninvasively.

Previous work shows that [15] KNN in random subsets may give accurate results with small datasets. Random forests are also used as ensemble learning classifiers to enhance the accuracy of single decision trees [16]. In this work, trained

models are evaluated using an independent test dataset. Results are then assessed within the context of the recent literature.

II. MATERIALS AND METHODS

A. Data sets

The patients' clinical data (WHO/ISUP grades included) and imaging data with semantic segmentation labels (See Fig. 1) were obtained from the Climb 4 Kidney Cancer-Kidney Tumor Segmentation Challenge (C4KC-KiTS) database [17], [18]. C4KC-KiTS database contained 210 RCC patients' data when the study was conducted. 143 of them identified as cc-RCC were used in this study.

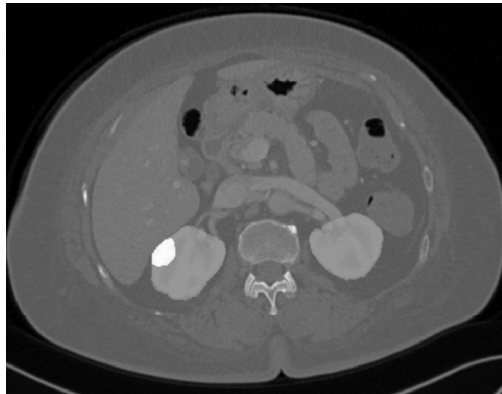


Fig. 1. cc-RCC patient with age 59, male, grade 2

B. Image pre-processing

Before radiomic feature extraction, all images were resampled and normalized. Digital Imaging and Communications in Medicine (DICOM) image format was used at all stages. Slice thicknesses were between 0.5 mm and 5 mm and pixel size was between 0.4375 mm and 1.041 mm before reconstruction and resampling processes. Therefore, all patients' images were resampled to obtain 1 mm $\times$ 1 mm $\times$ 1 mm spatial resolution because texture features need the same voxel size to be comparable. Resampling was performed using the SimpleITK image analysis toolkit [19] and Python software (version 3.7). Cubic B-Spline interpolation [20] method was used to find new values of resampled images. Intensity values of images were scaled using the Z-score normalization method [21].

C. Feature extraction

3D Slicer software [22] (version 4.10.2) was used for radiomic feature extraction. Before starting feature extraction with PyRadiomics [23] extension for 3D Slicer, gray level discretization was performed by adjusting the bin width to 0.01 in 3D slicer. Note that it is important to reduce influences of heterogeneity caused by acquisition and reconstruction protocols [24]. Features of the original images, Laplacian of Gaussian (Log) filtered images, and wavelet-transformed images were extracted. Laplacian of Gaussian filter values were chosen as 2 mm, 4 mm, and 6 mm to account for narrow, normal, and thick patterns. Radiomic analysis includes the conversion of images into minable data. Radiomic features are quantitative features which are calculated according to pixel intensities within a

ROI/VOI. With this approach, different aspects of ROI/VOI are quantified and phenotypic characteristics are represented quantitatively. After converting images into features, machine learning (ML) models are trained and tested by using the formed data table. Please refer to [23] for further details on the full description of features. For example, one of the radiomics features called Zone Entropy, which measures randomness in the distribution of zone sizes and gray levels, is calculated for each patient and allows researchers to compare images with numerical values.

D. Data pre-processing

20% of the data was reserved for testing and 80% was reserved for training. After stratified data splitting, test set included 29 instances (20 for low grade, 9 for high grade) and the training set included 114 instances (77 for low grade, 37 for high grade). The training set was spared for training and hyper-parameter optimization of the ML model while the test set was spared for the independent test of the final model.

E. Feature selection

Correlation coefficient analysis, Random Forest feature importance analysis and backward elimination methods were used consecutively [25]. Correlation coefficient analysis (CCA) was implemented first as a filter-based approach to eliminate redundant features that were very highly correlated to each other (absolute correlation coefficient > 0.9). Pearson's correlation coefficients were calculated for testing features correlations and collinearities. The RF algorithm was implemented with the Gini impurity which is a measure of how often a randomly chosen element from the set would be incorrectly labeled. Fifteen most valuable features were selected with the Sequential backward selection method (Logistic regression was used as an estimator). All of the feature selection methods were implemented in Python software using MLxtend library [26], Scikit-learn library [27], and Imbalanced-learn library [28].

F. Model training and evaluation

The ML-based classification was performed by using Random Forest and KNN with random subspace in the Classification learner app of Matlab R2019b software. Random subspaces-KNN and RF are among the most powerful ensemble ML algorithms. They are preferred in the presence of non-linear features and small sample sizes [15], [16], [29]. Model training was performed by using a training dataset with 154 instances augmented by the use of synthetic minority oversampling technique (SMOTE) [30] and cross-validation ($k=5$). Random subspace KNN models do not use all training data at once. Therefore, subspaces were created with random selection, multiple KNN models (ensemble models) were trained and combinations of these models were tested using cross-validation. Thousands of combinations were tried to find the best KNN model working with the highest accuracy. In the RF algorithm, a large number of uncorrelated small trees were trained to outperform the individual constituent models. Best hyperparameters (learner type, number of learners, subspace dimension and maximum number of splits) for RF and subspace-KNN algorithms were obtained. After finding the

best classification models, they were tested with the holdout test dataset. The validation and test results of the model were measured by calculating total accuracies and area under the curves (AUCs).

III. RESULTS

A. Data preprocessing and feature extraction

All images were fixed to $1 \text{ mm} \times 1 \text{ mm} \times 1 \text{ mm}$ spatial resolution. The total number of extracted radiomic features was 1133 and included 744 wavelet-transformed image features using eight different wavelet filters and six feature classes (first-order, gray level dependence matrix (gldm), gray level co-occurrence matrix (glcm), gray level run-length matrix (glrlm), gray level size zone matrix (glszm) and neighboring gray tone difference matrix (ngtdm)). Other features included 110 original and 279 features obtained from images filtered using the Laplacian of Gaussian (log) filter.

After applying the SMOTE algorithm, a new balanced training dataset with 77 samples for each grade, was generated (154 total). Two hundred and forty-two features were selected in the first feature selection process which was carried out by correlation coefficient analysis. 78 of 242 features were selected with the RF algorithm and 15 most valuable features were obtained with the Sequential backward selection method. The top four features were Inverse Difference Moment Normalized, Zone Entropy, Maximal Correlation Coefficient and Large Dependence High Gray Level Emphasis with importance ranging from 0.11 to 0.90.

B. Performance evaluation of ML models in the training set

The training set was used to construct classification models by using the Matlab Classification learner application. RF and subspace-KNN classifiers' results were obtained in the first run and hyperparameter tuning and optimization processes were conducted by using cross-validation ($k=5$) to find the best parameters of the classifiers. It took 5 hours for each training to optimize and find the best hyperparameters and validation accuracies and confidence intervals (CIs) were found as 86% (95% CI: 80-91.4) for RF and 90% (95% CI: 85.2-94.7) for subspace-KNN.

C. Performance evaluation on test data

The trained and optimized models were evaluated by using the test dataset and it took about 3 seconds to predict the grades. Testing accuracy of RF was 83% (95% CI: 70.1-96.0) and accuracy of KNN with random subspace 90% (95% CI: 80-100). The confusion matrices for the RF algorithm and random subspace-KNN algorithm on the test data are shown in Fig. 2. The ROC curves for both algorithms are shown in Fig. 3.

0	17	3	0	19	1
1	2	7	1	2	7
	0	1		0	1

Fig. 2. Confusion matrices for RF (on the left side) and random subspace-KNN (on the right side) on test dataset

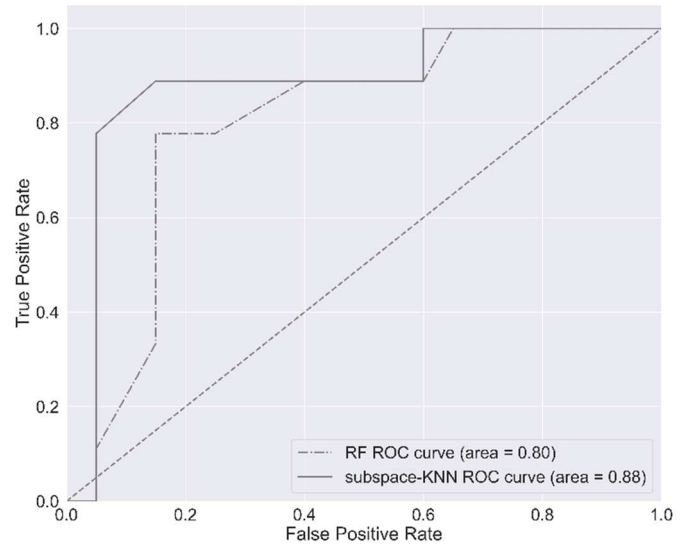


Fig. 3. ROC curves of classification models on the test dataset

Fig 3. shows that the random subspace-KNN ROC curve has an area under the curve (AUC) equal to 0.88 while the RF ROC AUC is equal to 0.80

IV. DISCUSSION

In this study, we explored whether ensemble learning based ML algorithms with radiomic analysis of CT images using 3D tumor volumes can be useful in predicting WHO/ISUP grades for cc-RCC patients using a small dataset. The results given in the previous section have shown the promising predictive performance of the ML algorithms using a reduced set of radiomic features. It was demonstrated that the random subspace-KNN algorithm performed better than the RF algorithm. However, it is important to point out that the used database is small and therefore larger datasets should be considered in the future in order to confirm this result. WHO/ISUP tumor grade is important in determining a patient's prognosis and developing a treatment plan. Therefore, this study may have a clinical impact as these grade estimates are obtained early and non-invasively.

This work differs from other reported work in the literature in some respects. The authors in this study have used only texture features to build an ML model, unlike Ding et al. [11], who used non-textural features. In previous studies, the 2-tiered prediction produced AUC of 0.67 in [14] and 0.86 in [13]. In the present study AUC was found 0.9 for the training cohort and 0.86 for the testing cohort (for KNN with random subspace) while pursuing 2-tiered classification

and 3D images were used to extract features. This is different than Ding et al. [11] who used only one slice (with largest tumor area) of CT images, and Tian et al. [12] who used 2D images. The logistic regression method was used to differentiate low-grades from high-grades by Ding et al. [11]. In our study, ensemble learning was used to improve accuracy.

Although our model with ensemble learning can predict grades with high accuracy and precision, further improvements may be explored by augmenting the features with clinical data and by using larger datasets.

V. CONCLUSION

We developed a WHO/ISUP grade prediction system with radiomic features that are extracted from 3D CT images of cc-RCC patients. Our independent test results show us that accurate tumor WHO/ISUP grading may be performed preoperatively and noninvasively by using a reduced set of texture features and ensemble learning based ML algorithms by using typically encountered small datasets.

ACKNOWLEDGMENT

We would like to thank the National Cancer Institute (NCI) and the Cancer Imaging Archive (TCIA) for providing the dataset.

REFERENCES

- [1] U. Capitanio and F. Montorsi, "Renal cancer", *The Lancet*, vol. 387, no. 10021, pp. 894-906, 2016.
- [2] V. Muglia and A. Prando, "Renal cell carcinoma: histological classification and correlation with imaging findings", *Radiologia Brasileira*, vol. 48, no. 3, pp. 166-174, 2015.
- [3] G. Novara, G. Martignoni, W. Artibani and V. Ficarra, "Grading Systems in Renal Cell Carcinoma", *Journal of Urology*, vol. 177, no. 2, pp. 430-436, 2007.
- [4] C. Lohse, M. Blute, H. Zincke, A. Weaver and J. Cheville, "Comparison of Standardized and Nonstandardized Nuclear Grade of Renal Cell Carcinoma to Predict Outcome Among 2,042 Patients", *American Journal of Clinical Pathology*, vol. 118, no. 6, pp. 877-886, 2002.
- [5] B. Delahunt, J. Cheville, G. Martignoni, P. Humphrey, C. Magi-Galluzzi, J. McKenney, L. Egevad, F. Algaba, H. Moch, D. Grignon, R. Montironi and J. Grigley, "The International Society of Urological Pathology (ISUP) Grading System for Renal Cell Carcinoma and Other Prognostic Parameters", *The American Journal of Surgical Pathology*, vol. 37, no. 10, pp. 1490-1504, 2013.
- [6] M. Ilić and P. Hofman, "Pros: Can tissue biopsy be replaced by liquid biopsy?", *Translational Lung Cancer Research*, vol. 5, no. 4, pp. 420-423, 2016.
- [7] W. Whittier, "Complications of the Percutaneous Kidney Biopsy", *Advances in Chronic Kidney Disease*, vol. 19, no. 3, pp. 179-187, 2012.
- [8] P. Lambin, E. Rios-Velazquez, R. Leijenaar, S. Carvalho, R. van Stiphout, P. Granton, C. Zegers, R. Gillies, R. Boellard, A. Dekker and H. Aerts, "Radiomics: Extracting more information from medical images using advanced feature analysis", *European Journal of Cancer*, vol. 48, no. 4, pp. 441-446, 2012.
- [9] V. Kumar, Y. Gu, S. Basu, A. Berglund, S. Eschrich, M. Schabath, K. Forster, H. Aerts, A. Dekker, D. Fenstermacher, D. Goldhof, L. Hall, P. Lambin, Y. Balagurunathan, R. Gatenby and R. Gillies, "Radiomics: the process and the challenges", *Magnetic Resonance Imaging*, vol. 30, no. 9, pp. 1234-1248, 2012.
- [10] R. Gatenby, O. Grove and R. Gillies, "Quantitative Imaging in Cancer Evolution and Ecology", *Radiology*, vol. 269, no. 1, pp. 8-14, 2013.
- [11] J. Ding, Z. Xing, Z. Jiang, J. Chen, L. Pan, J. Qiu and W. Xing, "CT-based radiomic model predicts high grade of clear cell renal cell carcinoma", *European Journal of Radiology*, vol. 103, pp. 51-56, 2018.
- [12] K. Tian, C. Rubadue, D. Lin, M. Veta, M. Pyle, H. Irshad and Y. Heng, "Automated clear cell renal carcinoma grade classification with prognostic significance", *PLOS ONE*, vol. 14, no. 10, p. e0222641, 2019.
- [13] C. Bektas, B. Kocak, A. Yardimci, M. Turkanoglu, U. Yucetas, S. Koca, C. Erdim and O. Kilickesmez, "Clear Cell Renal Cell Carcinoma: Machine Learning-Based Quantitative Computed Tomography Texture Analysis for Prediction of Fuhrman Nuclear Grade", *European Radiology*, vol. 29, no. 3, pp. 1153-1163, 2018.
- [14] Z. Feng, Q. Shen, Y. Li and Z. Hu, "CT texture analysis: a potential tool for predicting the Fuhrman grade of clear-cell renal carcinoma", *Cancer Imaging*, vol. 19, no. 1, 2019.
- [15] T. Ho, "Nearest neighbors in random subspaces." *Joint IAPR International Workshops on Statistical Techniques in Pattern Recognition (SPR) and Structural and Syntactic Pattern Recognition (SSPR)*, pp. 640-648, 1998.
- [16] L. Rokach, "Decision forest: Twenty years of research", *Information Fusion*, vol. 27, pp. 111-125, 2016.
- [17] N. Heller, et al. "Data from C4KC-KiTS" by The Cancer Imaging Archive, 2019. DOI: 10.7937/TCIA.2019.IX49E8NX
- [18] K. Clark, et al. "The Cancer Imaging Archive (TCIA): Maintaining and Operating a Public Information Repository", *Journal of Digital Imaging*, Volume 26, Number 6, pp. 1045-1057, 2013.
- [19] Z. Yaniv, B. Lowekamp, H. Johnson and R. Beare, "SimpleITK Image-Analysis Notebooks: a Collaborative Environment for Education and Reproducible Research", *Journal of Digital Imaging*, vol. 31, no. 3, pp. 290-303, 2017.
- [20] Z. Wang, K. Wang, S. An. "Cubic B-spline interpolation and realization", *International Conference on Information Computing and Applications*, pp. 82-89, 2011.
- [21] X. Cao, I. Stojkovic and Z. Obradovic, "A robust data scaling algorithm to improve classification accuracies in biomedical data", *BMC Bioinformatics*, vol. 17, no. 1, p. 359, 2016.
- [22] A. Fedorov, R. Beichel, J. Kalpathy-Cramer, J. Finet, J. Fillion-Robin, S. Pujol, C. Bauer, D. Jennings, F. Fennessy, M. Sonka, J. Buatti, S. Aylward, J. Miller, S. Pieper and R. Kikinis, "3D Slicer as an image computing platform for the Quantitative Imaging Network", *Magnetic Resonance Imaging*, vol. 30, no. 9, pp. 1323-1341, 2012.
- [23] J. van Griethuysen, A. Fedorov, C. Parmar, A. Hosny, N. Aucoin, V. Narayan, R. Beets-Tan, J. Fillion-Robin, S. Pieper and H. Aerts, "Computational Radiomics System to Decode the Radiographic Phenotype", *Cancer Research*, vol. 77, no. 21, pp. e104-e107, 2017.
- [24] R. Larue, J. van Timmeren, E. de Jong, G. Feliciani, R. Leijenaar, W. Schreurs, M. Sosef, F. Raat, F. van der Zande, M. Das, W. van Elmpt and P. Lambin, "Influence of gray level discretization on radiomic feature stability for different CT scanners, tube currents and slice thicknesses: a comprehensive phantom study", *Acta Oncologica*, vol. 56, no. 11, pp. 1544-1553, 2017.
- [25] I. Guyon, E. André. "An introduction to variable and feature selection", *Journal of machine learning research*, pp. 1157-1182, 2003.
- [26] S. Raschka, "MLxtend: Providing machine learning and data science utilities and extensions to Python's scientific computing stack", *Journal of Open Source Software*, vol. 3, no. 24, p. 638, 2018.
- [27] F. Pedregosa, et al. "Scikit-learn: Machine learning in Python." *The Journal of machine Learning research*, 12, pp. 2825-2830, 2011.
- [28] G. Lemaitre, F. Nogueira, C. Aridas, "Imbalanced-learn: A python toolbox to tackle the curse of imbalanced datasets in machine learning." *The Journal of Machine Learning Research*, 18(1), pp. 559-563, 2017.
- [29] T. Shaikhina, D. Lowe, S. Daga, D. Briggs, R. Higgins and N. Khovanova, "Decision tree and random forest models for outcome prediction in antibody incompatible kidney transplantation", *Biomedical Signal Processing and Control*, vol. 52, pp. 456-462, 2019.
- [30] N. Chawla, K. Bowyer, L. Hall and W. Kegelmeyer, "SMOTE: Synthetic Minority Over-sampling Technique", *Journal of Artificial Intelligence Research*, vol. 16, pp. 321-357, 2002.